The Runtime of Geometric Core Subset Sum

Jonathan R. Landers

October 28, 2026

Runtime companion to Note VII · Working draft

Purpose. Note VII [4] gives an exact geometric route and fresh paid-completion gains. This companion explains its runtime: discover a narrow core, build one exact table, and enumerate the exceptional items. The gain depends on the discovered structure. It supplies no improved general worst-case subset-sum bound.

The parameters that control the work

After exact reductions, let $u_1, \dots, u_n$ be positive integer weights, T the target, and L their total binary encoding length. Discover $u_i = Mp_i + r_i$, with $M \geq 2$ and $p_i \geq 0$, and split the indices into a core C of size m and exceptions E of size k. Define

$$R_- = \sum_{i \in C} \min(0, r_i), \quad R_+ = \sum_{i \in C} \max(0, r_i), \quad Q = \min\left(\sum_{i \in C} p_i, \left\lfloor \frac{T - R_-}{M} \right\rfloor\right).$$

For an eligible plan with $Q \geq 0$, the exact core table has dimensions

$$H = Q + 1, \qquad W = R_+ - R_- + 1.$$

The two coordinates are selected quotient sum and selected residual sum. For an exceptional subset A, query the shifted target $z_A = T - \sum_{i \in A} u_i$. A possible quotient must satisfy

$$\max\left(0, \left\lceil \frac{z_A - R_+}{M} \right\rceil\right) \leq q \leq \min\left(Q, \left\lfloor \frac{z_A - R_-}{M} \right\rfloor\right).$$

This interval contains at most $G = \min(H, \lfloor (W-1)/M \rfloor + 1)$ candidates. Test the corresponding exact pair $(q, z_A - Mq)$; interval feasibility alone cannot prove YES. All 2^k exceptional choices share the same table.

One construction plus exceptional queries

Let F pay reductions, acquisition and discovery; B pay enumeration and bounds per assignment; D pay one table construction; P pay one point test; and V pay reconstruction and verification. If I indicates that any assignment survives the necessary bounds, then

$$\boxed{T_{\text{route}} \leq F + 2^k B + ID + 2^k GP + V.}$$

Bounds-first builds only upon the first surviving assignment. For the same plan and sufficient resources, the isolated saving over eager construction is exactly $(1 - I)D$. If every assignment is excluded, it certifies NO without any table. Otherwise it still pays for one complete table, never one per assignment.

With word width ω, a conservative bitset estimate is $D = O((m+1)H(1 + \lceil W/\omega \rceil))$ word operations, apart from scalar arithmetic. Python shifts used for point tests also have width-dependent cost; P is not automatically constant. Prefix tables for witnesses occupy at most $(m+1)HW$ logical bits, plus row, pointer and interpreter overhead.

In the bit-cost model, suppressing polynomial factors in L with O^* gives the convenient upper bounds

$$T_{\text{route}} = F + O^*(HW + 2^k GW) \ \leq \ F + O^*(2^k HW).$$

The first expression exposes core reuse. The second is a looser shorthand, not a claim that the table is rebuilt 2^k times. Neither hides H, W or 2^k as polynomial input costs.

When the bound gives an advantage

Polynomial structured regime. If discovery costs $\mathrm{poly}(L)$, $HW = \mathrm{poly}(L)$, and $k = O(\log L)$, the complete route takes polynomial bit time. Indeed $G \le H$ and $2^k = \mathrm{poly}(L)$ bound every term above. Small k alone is insufficient: the numeric table dimensions can be exponential in binary input length. A useful core must also be found and checked within the discovery allowance. We do not prove this detector finds every affordable core.

Exponential-scale comparison. The coarse bound places the route below a benchmark scale 2^{cn} whenever, for some fixed $\epsilon > 0$,

$$F = O^*(2^{(c-\epsilon)n}), \qquad k + \log_2(HW) \le (c-\epsilon)n.$$

This is a sufficient condition, not a necessary one; the sharper bound above can do better. For the usual n-based comparison assume polynomial-bit inputs, so polynomial factors in L are subexponential in n. Use $c = 1/2$ for the classical scale, or $c = 0.49$ for the recent preprint's claimed scale. These inequalities compare bounds, not measured speedups or hidden constants.

General algorithms and the scope of the comparison

The table compares classical algorithms; n counts input integers. The cited randomized algorithms have bounded error, whereas our completed decisions are exact. O^* suppresses polynomial encoding-length factors; ordinary O in the cited results uses their stated RAM model.

Method	Time bound	Scope
Horowitz–Sahni [1]	$O^*(2^{n/2})$	General, deterministic exact search.
Chen et al. [2]	$O(2^{n/2}n^{-\gamma})$	Published randomized result; $\gamma > 0.5023$, word/circuit RAM.
October 2026 preprint [3]	Claims $O(2^{0.49n})$	General randomized decision; polynomial-bit inputs, stated word RAM.
Our geometric route	$F + O^*(HW + 2^k GW)$	Exact when completed; depends on discovered core and exceptions.

The October 4 preprint states success probability at least $2/3$ per input and its runtime cap on every random execution [3]. It is a recent claim, not an algorithm implemented or independently validated in our experiments. The published comparison remains useful because a polynomial-factor improvement and a smaller exponential rate are different claims. Target-dependent pseudopolynomial algorithms are another comparison axis: even elementary $O(nT)$ dynamic programming can be preferable for small numerical T; this is not polynomial in $\log T$.

What Note VII establishes

The implementation caps exceptions, table/history size, work and time. On exhaustion it may return *unresolved*, so bounded runtime does not make it a complete polynomial-time algorithm. A complete MITM fallback, preceded by a polynomially budgeted geometric attempt, would preserve $O^*(2^{n/2})$ worst-case time. This completion guarantee is a possible wrapper, not a new bound proved for the capped experimental portfolio.

Note VII reports 453 to 469 completions on 576 fresh targets for core reuse versus root geometry. The later bounds-first refinement avoids 90 of 182 tables; its primary incremental completion comparison is only 953 versus 954, with interval $[0, 3]$. These are paid-budget results, not an asymptotic improvement or a demonstrated speedup over the algorithms above. The research target is to enlarge the families where cheap discovery yields small k and compact HW, while preserving exact decisions and charging all acquisition costs. No new experiment is run for this companion.

References

[1] E. Horowitz and S. Sahni. Computing partitions with applications to the knapsack problem. *JACM* 21(2), 277–292, 1974. https://doi.org/10.1145/321812.321823.

[2] X. Chen, Y. Jin, T. Randolph and R. A. Servedio. Subset Sum in Time $2^{n/2}/\mathrm{poly}(n)$. *APPROX/RANDOM*, 39:1–39:18, 2023. https://doi.org/10.4230/LIPIcs.APPROX/RANDOM.2023.39.

[3] OpenAI. *Subset Sum in Time* $O(2^{0.49n})$. Preprint, October 4, 2026. Manuscript and stated computational model.

[4] J. R. Landers. *Geometric Cores and Exceptions in Subset Sum.* Note VII, working draft, 2026. Main note, proofs and frozen-study evidence.